Automated Passenger Boarding Kiosk

Problem definition, data sources and solution strategy

Problem

At the gate, an agent checks every passenger by hand: does the name on the ID match the boarding pass and the flight manifest, is the date of birth right, is this the person in the ID photo, and is the boarding pass for this flight, seat and time? It is slow, inconsistent between agents, and produces no record of what was checked. The airline also gets no structured feedback on how passengers experienced boarding.

The goal is a self-service **kiosk** that performs these checks automatically from the documents and a short video the passenger already provides, shows a clear message (board, or see an agent and why), and records every result against the **flight manifest**. A separate **luggage check** flags prohibited items (lighters) in carry-on luggage.

Data sources

Source	Content	Used for
Flight manifest (CSV)	One row per passenger: ticket number, name, date of birth, sex, carrier, flight, class, route, date, seat, gate, boarding time, and five validation statuses (all FALSE at start)	Ground truth every check is compared against
Digital ID (PNG, California driver licence)	Name, date of birth, sex, face photo	Name, DoB and Person Validation
Boarding pass (PDF)	Name, carrier, flight, class, route, date, seat, gate, boarding time, ticket number	Manifest lookup, Name and Boarding Pass Validation
Identity video (MP4, ~30 s)	The passenger's face; any speech	Person Validation, passenger feedback
Luggage scan images (JPG)	Carry-on items	Luggage Validation (lighter detection)
Lighter training images (47 JPG)	Photos of lighters	Training the lighter detector

Solution strategy and Azure services

Input	Azure service	What it produces
Digital ID	Document Intelligence (formerly Form Recognizer), prebuilt `idDocument` model	First/last name, date of birth, sex
Boarding pass	Document Intelligence custom **template** model, trained on 12 synthetic passes generated from the airline template and labeled with 10 fields	All flight fields and the ticket number
Identity video	Video Indexer (successor of Video Analyzer)	Face thumbnails, emotions, sentiment

Video faces + ID photo	Face API: a person model trained on the video face frames, then *identify* with the ID face; *verify* as a cross-check	Face-match confidence
Luggage images	Custom Vision object detection, tag `lighter`	Lighter probability per image
Everything	Blob Storage (7 containers: manifest, IDs, passes, training passes + labels, videos, luggage, results)	Inputs, training data, per-passenger results, final manifest

The kiosk looks the passenger up on the manifest by the **ticket number** on the boarding pass (not by name), runs the five validations, writes TRUE/FALSE into the manifest and chooses the kiosk message.

Data validation, metrics and thresholds

Validation	Rule	Metric / threshold
Name (3-way)	Boarding pass name and ID name both match the manifest; case, punctuation and middle initials ignored (first given name + last surname word)	Exact after normalization; extraction confidence logged
Date of birth	ID date equals manifest date as a calendar date	Exact
Boarding pass	Flight number (carrier + number), seat, class, origin, destination, date, boarding time all equal the manifest	Exact after normalization; model's estimated field accuracy ≥ 0.95 on training
Person	ID face matches the face in the video	Face confidence **≥ 0.65**
Luggage	No lighter in the luggage scan	Max lighter probability **< 0.5**; model precision must be **≥ 75%**

Unreadable evidence (a field the model could not read, an unparseable date) makes the validation fail. It is never skipped. Message order: a boarding-pass failure is shown first, then an ID failure (name/DoB). Otherwise the passenger is welcomed with flight, time, route and seat, followed by one luggage sentence and one identity sentence. A passenger whose luggage was not scanned is not told a lighter was found.

Passenger feedback (emotions and sentiment from the video) is stored as a positive/negative/neutral label with each passenger's result. It is informational and does not affect boarding.